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Computing and Python Programming Review Worksheet

Part 1: Key Definitions & Vocabulary

Review these essential terms before starting the worksheet activities.

- **Raster Graphics:** Images made of a fixed grid of tiny colored dots called **pixels** (e.g., PNG, JPEG). When enlarged, they lose quality and look blurry or pixelated.
- **Vector Graphics:** Images built using mathematical formulas defining lines, shapes, and curves (e.g., SVG). They can be scaled infinitely to any size without losing sharpness.
- **Raster Transparency:** The ability of a raster image format (like PNG) to support see-through or semi-transparent backgrounds using an **alpha channel**. Unlike JPEG (which fills empty spaces with solid white or black), a transparent PNG allows background colors or other images to show through its empty pixels.
- **Selection:** A programming concept where the computer decides which path or block of code to run based on whether a specific condition is true or false (e.g., using if / else statements).
- **Interpreting:** The process of translating and executing code line-by-line at runtime using an interpreter program.
- **Compiling:** The process of translating an entire program's source code into a binary machine-code file all at once before the program can run.
- **Vibe Coding:** A modern, high-level development style where a developer describes their software goals to an AI using natural language and directs its generation, focusing on architecture and system design rather than typing individual lines of code.

Part 2: Key Computing Concepts (Study & Review)

Read through these concepts to prepare for the questions below.

1. File Formats: PNG vs. SVG

- **PNG (Portable Network Graphics):** A raster image format supporting raster transparency.

When to use: Best for complex images, photographs, or graphics with detailed color gradients.

Limitations: Enlarging a PNG will cause pixelation.

- **SVG (Scalable Vector Graphics):** A vector image format.

When to use: Best for logos, icons, illustrations, and user interface elements.

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Benefits: Scales infinitely without quality loss, and file sizes are typically very small.

2. Cross-Device Compatibility: Single HTML Files

A single, self-contained HTML file (containing HTML structure, CSS styling, and JavaScript logic) is highly portable.

Why it works everywhere: Modern web browsers on every operating system (Windows, macOS, iOS, Android, Linux) universally support these standard web technologies. Because the code is self-contained, the file does not need an active internet connection to run, it can be opened directly on almost any device.

3. "Vibe Coding" and Debugging with AI

Vibe Coding: The developer acts as a systems director or conductor, using natural language to prompt AI assistants to write and assemble code.

AI Debugging: Developers can share error logs or unexpected code behaviors with AI models to quickly diagnose, explain, and correct syntax or logical errors.

4. Interpreted vs. Compiled Languages

Interpreted Languages (e.g., Python, JavaScript): Code is executed line-by-line by an interpreter at runtime.

Pros: Faster to test, modify, and run immediately.

Cons: Slower execution speeds.

Compiled Languages (e.g., C++, Rust, Go): Code is translated into a machine-readable binary executable before running.

Pros: Fast execution speed.

Cons: Requires a compiling step whenever changes are made, and compiled files must target specific operating systems.

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Part 3: Multiple Choice Questions

Circle the correct letter for each question.

1. Which of the following best describes "selection" in computer programming?

- A. Choosing which programming language to use.
- B. Running a specific block of code only if a certain condition is met.
- C. Repeating a block of code a set number of times.
- D. Selecting text in an IDE to copy and paste.

2. Which Python keyword is used to check an additional condition if the initial if statement is false?

- A. else
- B. or
- C. elif
- D. if-else

3. What does "raster transparency" allow a PNG image to do?

- A. Scale to any size without losing quality.
- B. Run embedded JavaScript code in a browser.

C. Allow background colors or other images to show through empty or semi-transparent pixel areas.

D. Convert automatically into a vector SVG file.

4. What does the `int()` function do in the line of code: `score = int(input("Enter your score:"))`?

- A. It prints the score to the screen.
- B. It stores the input as a decimal (float).
- C. It converts the user's input from a string of text into an integer (whole number).
- D. It automatically checks if the score is greater than 80.

5. Which type of language is translated entirely into machine code before execution, resulting in faster performance?

- A. Interpreted language
- B. Markup language
- C. Compiled language
- D. AI-generated language

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Part 4: Gap-Fill Review

Fill in the blanks using the word bank below. (Some words may not be used, and some may be used more than once.)

Word Bank:

if | elif | else | input | int | compiled | interpreted | SVG | PNG | selection | browsers | pixels | vectors

1. In Python, we program _____ using the keywords if, elif, and else to control the flow of our program.

2. Fill in the gaps in this code snippet from the *Pyphone Passcode* program:

```
passcode = _____("Enter your passcode:")
```

```
if passcode == "1234":
```

```
    print("Welcome")
```

```
_____:
```

```
    print("Incorrect passcode")
```

3. Python is considered an _____ language because its code is executed line-by-line at runtime, unlike C++ which is a _____ language.
4. A single HTML file can run on any device because web _____ on different operating systems universally support standard HTML, CSS, and JavaScript.
5. _____ images are composed of mathematical formulas, allowing them to scale without loss of quality, whereas _____ images are composed of fixed grids of pixels.

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Part 5: Code Analysis & Short Answer

Based on the *Grades Calculator* program:

```
score = int(input("Enter your score: "))

if score >= 80:

    print("Grade A")

elif score >= 70:

    print("Grade B")

elif score >= 60:

    print("Grade C")

else:

    print("Oh dear!")
```

1. What will the program output if the user enters 75?

2. What will the program output if the user enters 55? Explain why.
